

Transformer OLTC Operation Monitoring Framework Through Acoustic Signal Processing and Convolutional Neural Networks

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Abstract—The increasing penetration of renewable energy sources (RESs) has a direct impact on the efficiency and reliability of the power grid. In this context, maintaining voltage stability and power quality has become more important than ever. Transformers are crucial components in the grid and are generally equipped with on-load tap changer (OLTC) mechanisms. With the OLTC, transformers can automatically adapt to different voltage levels depending on the dynamics of the grid. Thereby, they assist in the penetration of RESs and improve system operation efficiency. However, the continuous grid fluctuations require increased activity from OLTCs, which, in turn, contributes to elevated stress on equipment and accelerated aging. Existing OLTC monitoring solutions are generally invasive and time-consuming. Accordingly, this research work presents a new cost-effective OLTC monitoring approach based on audio signals. The approach is inspired by self-paced learning (SPL) strategies. Initially, different monitoring solutions are evaluated and tested with respect to accuracy, computational cost, and robustness to noise. After the best architecture is selected, the audio signal processing approach is further refined through signal processing and convolutional neural networks (CNNs). The proposed approach is benchmarked against automated machine learning (AutoML), which shows a competitive performance, but with increased computational cost. The approach has been validated using real data from a transformer in operation.

Index Terms—Audio, machine learning, on-load tap changer (OLTC), signal processing, transformer.

I. INTRODUCTION

A RESILIENT power and energy infrastructure is crucial to transit toward more sustainable energy production. However, increasing penetration of renewable energy sources

(RESs) has a direct impact on the efficiency and reliability of the power grid. In this context, elementary power assets, such as power transformers, have been upgraded with functionalities to assist with the penetration of RESs and improve power system operation efficiency [1]. As maintaining voltage stability and power quality has become more important than ever, power transformers equipped with on-load tap changer (OLTC) mechanisms are becoming increasingly widespread. OLTC transformers can automatically adapt to different voltage levels according to the power grid dynamics, and in this context, there are emerging control solutions that minimize voltage deviations from the setpoint [2]. However, the continuous fluctuations in the power network necessitate increased activity from these devices, which, in turn, contributes to elevated stress on the equipment. The OLTC is an electromechanical system, which at least includes a motor, a tap selector, and a diverter switch that performs the tap-change operation. Due to the operation regimes of the OLTC, its constituent components induce different types of failures [3]. However, the run-to-failure trajectory of OLTC components is difficult to monitor, as it requires monitoring failure precursors over long periods of time, which is not always feasible and cost-effective. On the contrary, continuous monitoring of the operational status of the OLTC is an interesting technological solution, as it will enable detecting deviations from normal operation and raising alarms. Repeated OLTC operation accelerates aging due to mechanical wear and electrical stress, particularly affecting contacts and switching mechanisms.

Consequently, this research focuses on noninvasive and cost-effective solutions to monitor the operational status of the OLTC system through audio signals. The study differs from conventional methods by utilizing audio signals rather than relying on vibration-based diagnostics with accelerometers, which require physical contact with the transformer. This approach offers greater flexibility, faster setup, and broader application using commercially available equipment. Predictive modeling of OLTC states is challenging, as audio signatures need to be correlated with specific states, including motor activation, diverter switch operation, and various idle states.

A. Related Work

Acoustic signals emitted during OLTC operation provide a rich source of diagnostic data. These signals can be captured

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using contact-based methods, such as accelerometers, or non-contact techniques, such as microphones. Compared to other diagnostic approaches, acoustic methods are advantageous as they enable continuous monitoring without interrupting the transformer's regular operation. By characterizing, monitoring, and comparing the operation of multiple OLTCs of the same type, deviations in performance can be identified early, improving fault detection and enhancing transformer reliability.

In early works, detecting individual acoustic sources was a key challenge. Given the mechanical nature of the OLTC system, high-speed cameras were utilized to identify discrete acoustic events. The total switching time of an OLTC typically spans several seconds, with the majority of this period consumed by the tap selector operation. Acoustic emissions recorded during this process were analyzed, and simple methods of comparing amplitude and timing deviations (on the order of a few milliseconds) proved to be effective for fault detection in the tap selector mechanism.

Greater attention in acoustic diagnostics has been given to the diverter switch, a critical component of the OLTC, which is responsible for interrupting current within milliseconds. This component experiences significant mechanical and electrical stress during operation, making its diagnostics particularly challenging. Detecting faults through time-domain analysis of signals emitted by the diverter switch is complex due to difficulties in proper signal alignment. The small magnitude of deviations in these signals requires precise diagnostic techniques. Several approaches have been utilized, including dynamic time warping (DTW) for accurate signal synchronization, signal simplification through envelope extraction, and converting signals into images for easier comparison using convolutional neural networks (CNNs). Diagnostics based on audio signals present unique challenges. While there is a strong correlation between vibration and audio patterns, the background noise from an active transformer poses significant difficulties. Deep neural networks (DNNs) have proven to be effective in mitigating this noise, significantly improving the reliability of audio-based diagnostic methods. Table I reviews the main OLTC diagnostics applications using ML and signal processing strategies, highlighting the evolution from basic acoustic methods to advanced machine learning techniques for early fault detection and improved transformer reliability.

In a broader application area, different solutions have been proposed. For example, Ding et al. [11] introduced a novel time-frequency transformer method based on the vanilla transformer architecture for fault diagnosis in rolling bearings. The model surpassed traditional approaches such as DNN, CNN, and recurrent neural networks (RNNs), achieving better accuracy and faster training speeds across different datasets. Dai et al. [12] introduced the acceleration-guided acoustic signal denoising framework for feature extraction from acoustic signals focused on monitoring slab track conditions. This method effectively classifies signals into three groups of slab track conditions and demonstrates resilience to noise and variations in train speeds. For a more detailed

exploration of fault diagnosis using audio signal analysis, see [13].

It can be observed that it is possible to improve transformer OLTC monitoring practice through the appropriate combination of signal processing methods and advanced ML strategies. Unlike reviewed diagnostic methods that learn classification criteria in a direct pass, self-paced learning (SPL) methods focus on gradual learning [14], [15] (see Appendix A for more details). That is, for diagnostic tasks, the coarse classification criteria are first learned, and then, it is refined in subsequent learning iterations. Consequently, the objective of this work is to develop an efficient SPL-inspired OLTC diagnostic approach through the use of acoustic signals, without additional sensing equipment.

Automated machine learning (AutoML) builds end-to-end ML pipelines for the automatic design and deployment of the combination of ML models. AutoML has been successfully applied to classification [16] and regression problems, such as predicting the remaining useful life (RUL) of assets [17]. The main benefit is that it alleviates the task of building hand-crafted predictive models, however, at the cost of predefining the problem and fixing constraints on training data. Given the potential of the AutoML methodology to create fine-tuned ensemble models, an AutoML-based classification model will be used as a benchmark strategy to compare the results obtained with the proposed approach.

B. Contributions and Organization

In this context, the main contribution of this research work is the proposal of an OLTC monitoring framework based on audio signals through the combination of CNN and signal processing techniques. The approach is designed through SPL strategies. First, a set of signal processing and classification algorithms are evaluated in terms of accuracy, complexity, and robustness to noise. Subsequently, after selecting the best architecture that learns coarse classification criteria, it is further refined through additional signal processing and learning strategies.

In addition, labeling challenges, computational efficiency strategies, balancing methods, and different testing strategies are evaluated and discussed. The proposed approach shows that it effectively captures the diverse states of the OLTC, underscoring its potential to enhance diagnostic processes.

The proposed approach presents a cost-effective and time-saving alternative to traditional vibration-based OLTC monitoring. In addition, it assists in the: 1) reduction of memory consumption by allowing the elimination of data during periods when the OLTC is not active and 2) identification and extraction of important features.

The labeled audio datasets, the source code, and extra materials, including detailed explanations of the experimental setup, microphone placement strategies, and additional visualizations of collected signals, are available at [18]. These supplementary materials provide comprehensive support for further exploration.

The remainder of this manuscript is organized as follows. Section II introduces the analyzed OLTC monitoring problem.

TABLE I
SUMMARY OF OLTC DIAGNOSTICS METHODS

Ref.	Data	Methods	Diagnostic Decision	Main Contribution
[4]	Vibration	DWT	Amplitude peaks, Euclidean distance	Experimentally demonstrated the detection of tap-selector faults through acoustic events and linked acoustic events to mechanical events using high-speed video camera.
[5]	Vibration and current	CNN	SVM	Utilized vibration and motor current signals to identify faults such as transmission shaft gear issues and contact looseness.
[6]	Audio	STFT/DNN	Amplitude peaks, Euclidean distance	Elimination of transformer-induced noise to retain clean audio diagnostic signals.
[7]	Vibration	DTW	Euclidean distance to reference signal	Vibration signal alignment to improve the comparison with reference signals for fault localization.
[8]	Vibration	Envelope extraction	Amplitude peaks, Euclidean distance	Application of vibro-acoustic diagnostic methods to long-term anomaly detection and monitoring.
[9]	Vibration	FSF/VFCAE	LPHHO	Full vibration signal analysis for OLTC gear fault detection.
[10]	Vibration	Envelope extraction	Euclidean distances	Diagnosis of contact wear in OLTCs through vibro-acoustic analysis and linear model

Legend: IMF: Intrinsic Mode Function, DWT: Discrete Wavelet Transform, STFT: Short-Time Fourier Transform, DTW: Dynamic Time Warping, DNN: Deep Neural Network, CNN: Convolutional Neural Network, SVM: Support Vector Machine, FSF: Frequency Statistical Feature, VFCAE: Vibration Frequency Component Amplitude Entropy, LPHHO: Limited-Patience Harris Hawks Optimization.

Section III presents the proposed approach. Section IV describes the case study. Section V evaluates the experimental results. Section VI introduces an extensive discussion about the approach and the results, and finally, Section VII concludes this article.

II. PROBLEM STATEMENT

The OLTC operation process encompasses the following seven stages [4].

- 1) The startup of the motor drive, corresponding to the first burst, initiates the OLTC transition. This phase is marked by inrush current and motor vibrations that form the initial acoustic profile.
- 2) Following this, the activation of the Geneva drive mechanism occurs, aligning with the second burst, when the position selector contact attached to the previous tap starts to move, creating distinctive sounds.
- 3) As the process continues, the moving contacts of the tap selector reach the final tap, generating the third burst of sound.
- 4) This is followed by the end of the movement of the Geneva drive, creating another distinct acoustic signature.
- 5) The operation of the diverter switch represents another phase, producing a series of bursts (the fifth in sequence) and generating an acoustic fingerprint.
- 6) The process concludes with the motor drive stopping, corresponding to the sixth burst.
- 7) After the transition, the OLTC enters an idle state in its new position, stopping to emit more sound, which signals the end of these diagnostic acoustic events.

Audio-based diagnostics offer a noninvasive alternative to traditional vibrational analysis, requiring no physical contact, which allows for faster and more flexible testing procedures. This method capitalizes on the availability of commercial sound recording systems, avoiding the need for

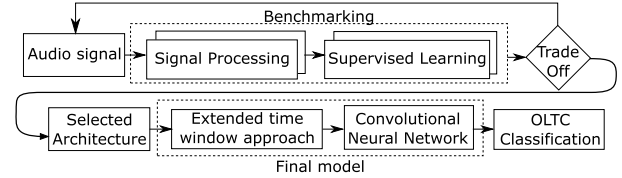


Fig. 1. Proposed OLTC operation monitoring approach.

specialized equipment. It is effective for the quick identification of problems, thereby guiding more focused diagnostic efforts.

Considering the desire for rapid execution in this type of diagnostics, it is not advisable to slow down or complicate the process with additional signals. Automatic classification of signals is the initial step that facilitates subsequent analysis and the comparison of signals corresponding to the OLTC transitions from one tap position to another. Considering that the OLTC spends most of its operational time in an idle state, where it emits no significant diagnostic information, automatically classifying and discarding signals corresponding to this state can also greatly reduce memory requirements [3].

III. PROPOSED APPROACH

The proposed OLTC classification approach is inspired by the SPL process, i.e., first learn coarse classification groups and then gradually learn ambiguous samples. The first objective is to determine the best combination of signal processing and classification methods for OLTC monitoring. Consequently, different signal processing and supervised classifiers have been designed and tested through a benchmarking exercise. The selection of the best configuration is based on accuracy, robustness to noise, and complexity of the model. After selecting the best architecture, a fine-tuning process is developed through the implementation of an additional signal processing algorithm along with the best classification architecture. Fig. 1 shows the proposed OLTC operation monitoring approach.

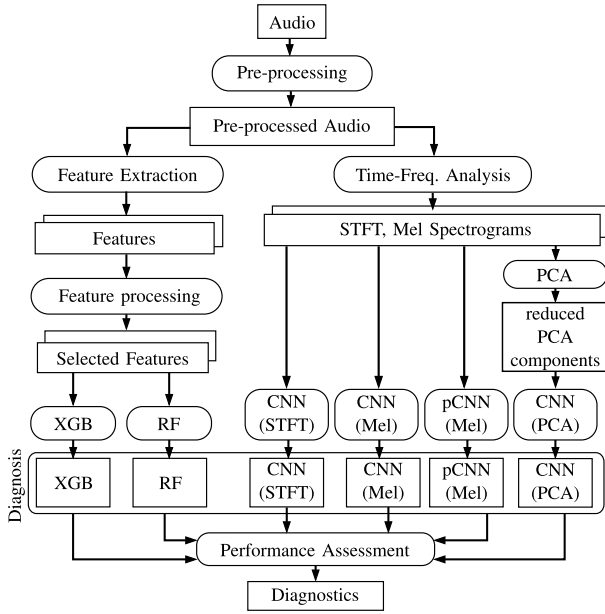


Fig. 2. Outline of the implemented benchmarking strategy.

A. Benchmarking Models

Two types of modeling strategies have been developed and validated. On the one hand, CNN-based feature map extraction and supervised learning strategies based on: 1) time–frequency signal transformations through short-time Fourier transform (STFT) and Mel spectrograms and 2) principal component analysis (PCA)-based feature extraction of the STFT spectrogram. CNNs combine the learning capability of classical neural networks (NNs) with the feature extraction ability of signal-processing methods through the filter optimization [19]. CNNs are particularly well-suited to grid-like topologies (e.g., images), as they slide along input features through filters (e.g., low-pass, bandpass, or high-pass) and provide a feature map. In addition, CNNs focus on dominant features and ignore minor random perturbations such as shifts or rotations of input data, which makes them robust to noise [20]. In addition, CNNs combined with Monte Carlo dropout strategies significantly improve the robustness to noise and provide a probabilistic interpretation of predictions [21]. In this direction, probabilistic CNN versions have been also designed to capture model uncertainty and evaluate the effect on diagnostics [22]. Accordingly, CNN models have been selected because they have shown an excellent ability to capture time–frequency information from signals for monitoring and diagnostics [23], [24]. On the other hand, classical feature extraction and learning approaches through statistical and frequency-based features are tested on tree-based classifiers focusing on random forests (RFs) and extreme gradient boosting (XGB) algorithms [25].

For all the configurations, the training dataset has been normalized to the sample mean 0 and variance 1. To test and validate the model, different train–test configurations have been designed and tested. Fig. 2 shows the overall outline of the implemented benchmarking strategy.

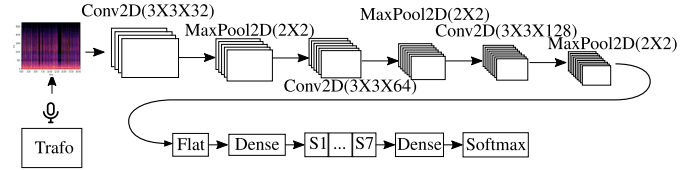


Fig. 3. Schematics of the CNN architecture. The numbers $a \times b \times c$ inside the Conv2D boxes indicate the kernel size and the number of filters.

1) *Time–Frequency Analysis-Based CNN*: The proposed CNN architecture for OLTC health state diagnostics is shown in Fig. 3.

The depth of an NN model can limit the probabilistic predictive capacity by producing overly confident predictions [26]. Due to the imbalance between classes, this may also worsen the calibration [27]. Accordingly, the CNN model with an eight-layer architecture has been designed, including convolutional filters, swish activation functions, and dense networks. For all the layers, 3×3 convolutional kernels are used. The number of filters for the first layer is 32, for the second layer 64, and for the last layer 128. Two max-pool operations are inserted between the convolutional layers after the activation functions. The output of the third layer is flattened and used as input of a dense layer with 54 neurons and a layer with seven outputs corresponding to the OLTC diagnostic states ($K = \{c_0, \dots, c_6\}$).

Classification states are then connected to a softmax activation layer, $f: \mathcal{R}^K \rightarrow (0, 1)^K$, which maps outputs into K classes, through normalized exponentiation for the probabilistic interpretation of results. The probability of the sample y corresponding to the class i is defined as [28]

$$f(y)_i = \frac{e^{y_i}}{\sum_{c=1}^K e^{y_c}}. \quad (1)$$

Finally, the loss function is defined through the sparse categorical cross-entropy [28]

$$\mathcal{L}_{CE}(W_{CE}) = - \sum_{i=1}^N \sum_{c=1}^K y_{c,i} \log(f(y)_{c,i}) \quad (2)$$

where W_{CE} corresponds to the weights of the CNN adjusted using the minimization of the cross-entropy loss and $y_{c,i}$ is the true label corresponding to the i th sample, and class c . The Adam optimizer has been used with a learning rate of 0.001, $\beta_1 = 0.9$, and $\beta_2 = 0.99$.

This is the base CNN model that has been used for training and testing, named CNN.

In order to select the most appropriate input data preprocessing stage for the CNN model, different time–frequency transformations have been applied to the audio signals, including STFT and Mel spectrograms.

2) *Probabilistic CNN*: Due to the inherent noise of the captured audio signals, a probabilistic version of the CNN model has been developed. The goal of the probabilistic CNN is to capture uncertainty associated with the prediction and adopt better-informed diagnostics decisions.

This has been achieved through the dropout mechanism, which can operate as a probabilistic Bayesian approach [22]. During the model training phase, the dropout layer prevents overfitting by randomly dropping out neurons. At testing time,

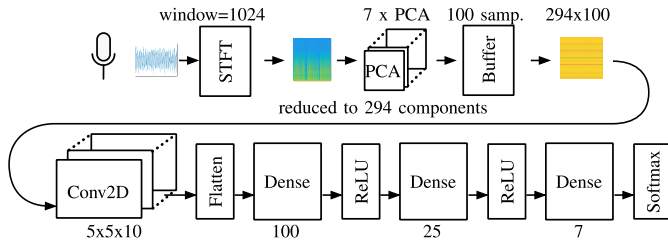


Fig. 4. PCA and CNN processing.

the repeated dropout execution enables the inference of a probability distribution, which captures model uncertainty.

Accordingly, the probabilistic diagnostics approach is based on Monte Carlo trials on different CNN models designed through the dropout mechanism. This configuration has been named pCNN. The pCNN model includes a dropout layer after each convolutional and fully connected layer in Fig. 3, with a dropout rate of 0.1.

After the evaluation of the learning and generalization capabilities of the CNN model with respect to input transformations, the pCNN model has been implemented with the most accurate time–frequency transformation.

3) *PCA-Based CNN*: In the case of PCA analysis-based input data, the captured input sound data for testing are first analyzed using STFT and processed to obtain the main components of each of the different transformer states. The dimension number of each of the PCA is reduced to cover 95% of the cumulative explained variance and not more than 50 dimensions in order to reduce the dimensionality of the input CNN. Finally, the obtained PCA vectors for the different states are concatenated to be used during data processing.

Data input to the CNN is processed similarly, as shown in Fig. 4, first processing the signal STFT, which is buffered, and the PCA analysis is performed. The output is buffered taking 100 time samples and introduced into the CNN as input.

4) *Feature Selection-Based Classification*: Using the audio signal, a set of features are extracted, which may help in the health state diagnostics. These features should be able to capture the OLTC dynamics to use them to train the classification model. Feature extraction is performed using a rolling window of length M ($M = 100$, in this case) for the available variables. Namely, the following statistical features are extracted: mean, maximum, minimum, kurtosis, crest factor, form factor, impulse factor, integral, derivative of the original signal, and frequency decomposition methods, including empirical mode decomposition.

The calculated features are then evaluated for their value in improving the classification accuracy. This is done in this case through the intrinsic importance functions of the ML models implemented below. The resulting variables with the highest predictive information are used in the testing set for validation purposes. For the diagnostics process, different ML models have been trained and tested.

RF is an ensemble of recursive trees [29]. Each tree is generated from a bootstrapped sample, and a random subset of descriptors is used at the branching of each node in the tree. RF creates a large number of trees by repeatedly resampling training data and averaging differences through

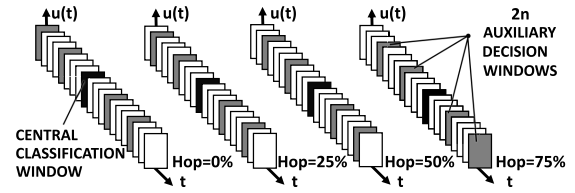


Fig. 5. Extended time window strategy example.

voting. RF is a valuable option for capturing the intricate relationships between the features due to its effectiveness in capturing complex relationships and handling nonlinearities.

XGB is a method that sequentially combines an ensemble of decision trees as a weighted sum that can be expressed as [30]

$$\hat{y} = \sum_{n=1}^N c_n f_n(x) \quad (3)$$

where N is the number of stages, $\hat{y}$ is the prediction variable, and each $f_n(x)$ is a regression tree or forest, which represents the steps to the final solution, with weight c_n , being x the set of values. Thus, it places greater emphasis on data points that were previously misclassified, leading to improved prediction accuracy. Therefore, given its robust performance, XGB proves to be a suitable choice for the proposed approach [30].

B. Final Model: Extended Time Window and CNNs

After learning coarse classification groups with the best benchmarking model, the aim of this stage is to gradually learn ambiguous samples, which is more effective than training all samples at once [14]. To this end, an extended time-window algorithm is developed.

The preprocessing phase is initiated by computing the Mel spectrogram, where M Mel coefficients are calculated, using w time windows, with a o overlap, which results in a time–frequency data matrix, $D = \{d_{1,1}, \dots, d_{L,M}\}$, with L Mel spectrogram windows and M Mel frequency bins.

The time–frequency matrix, D , is then analyzed to classify the spectral content into predefined OLTC states. In the proposed extended time window approach, n nonoverlapping auxiliary windows are selected around a central classification window, C , located before and after the central window, with an in-between window distance of d , i.e., a total of $n \times 2$ nonoverlapping windows are used, $n \in \mathbb{N}^+$.

This approach ensures the integration of wider temporal information along with the frequency content, rather than focusing on detailed transitional spectral analysis. Algorithm 1 outlines the proposed approach, which guides the CNN by dividing time–frequency matrices and passing them through convolutional layers. The CNN processes these segments and extracts features from the OLTC signals. This interaction allows the CNN to gradually improve the classification criteria, learning low- and high-frequency patterns in the signals. The algorithm's iterative approach improves the accuracy of the model over time by refining the model through uncertain cases. Note that the `mode` variable denotes the arrangement of windows, which, in this case, is horizontal.

Fig. 5 shows an example of the proposed time window approach for $n = 2$, $d = 4$, and four sequential iterations. The

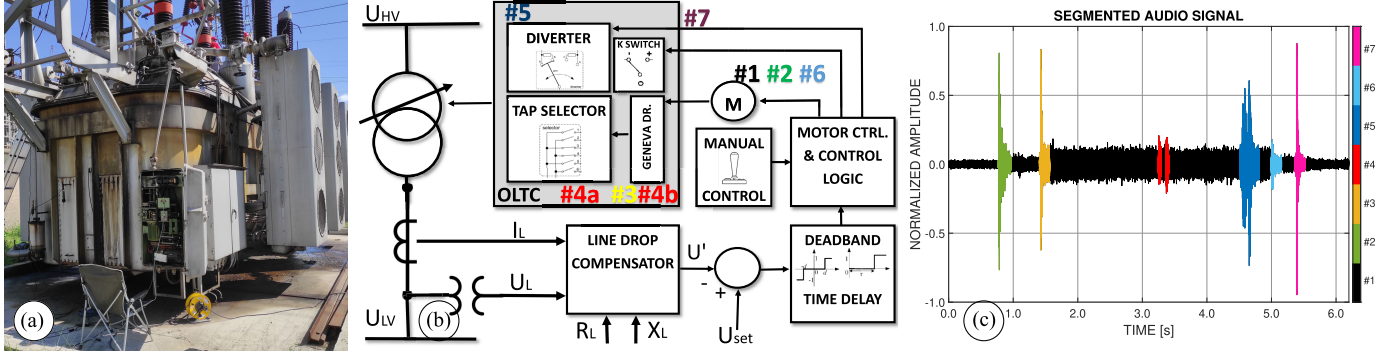


Fig. 6. (a) ELIN test environment. (b) Control scheme—acoustic sources. (c) Segmented audio signal.

Algorithm 1 Extended Time Window Approach

Inputs:

$D = [d_{1,1}, \dots, d_{L,M}]$ $\triangleright$ Data vector of $L \times M$ elements.

C $\triangleright$ Central classification window

n $\triangleright$ Auxiliary decision windows

d $\triangleright$ Distance between selected windows

$\mathcal{C}$ $\triangleright$ CNN architecture

$\mathcal{O}$ $\triangleright$ Optimizer

mode $\triangleright$ Window arrangement mode

Output: $\mathcal{V}$ $\triangleright$ Classification predictions.

Procedure:

```

1:  $i_c = \text{index}(C)$   $\triangleright$  get central classification window index
2: if mode='horizontal' then  $\triangleright$  horizontal alignment
3:    $\mathcal{W}_b = \{W_{i_c-d \cdot (k+1)}\}_{k=0}^{n-1}$   $\triangleright$  Before
4:    $\mathcal{W}_a = \{W_{i_c+d \cdot (k+1)}\}_{k=0}^{n-1}$   $\triangleright$  After
5:    $\mathcal{W} = \mathcal{W}_b \cup C \cup \mathcal{W}_a$   $\triangleright$  Create  $\mathcal{W}$  windows
6: else if mode='vertical' then  $\triangleright$  1D vertical alignment
7:    $\mathcal{W}_b = \bigoplus_{k=0}^{n-1} W_{i_c-d \cdot (k+1)}$   $\triangleright$  Concatenate before
8:    $\mathcal{W}_a = \bigoplus_{k=0}^{n-1} W_{i_c+d \cdot (k+1)}$   $\triangleright$  Concatenate after
9:    $\mathcal{W} = \mathcal{W}_b \oplus C \oplus \mathcal{W}_a$   $\triangleright$  Concatenate with central
10: end if
11:  $\mathcal{W}' = (\mathcal{W} - \mu) / \sigma$   $\triangleright$  z-score normalization
12:  $[\mathcal{W}_{Tr}, \mathcal{W}_{Te}] = \text{split}(\mathcal{W}')$   $\triangleright$  split into train and test sets
13:  $\mathcal{C} = \text{Initialize}(\mathcal{C}, \vec{\Theta})$   $\triangleright$  Initialize CNN model with weights
14: for each epoch  $e = 1$  to  $E$  do
15:   for each batch  $b \in \mathcal{W}_{Tr}$  do
16:      $\hat{y} = \mathcal{C}(b, \vec{\Theta})$   $\triangleright$  predictions, forward propagation
17:      $l_{CE} = \mathcal{L}_{CE}(\hat{y}, y)$   $\triangleright$  Compute loss [Eq. (2)]
18:      $\vec{\Theta} = \mathcal{O}(\mathcal{C}, l_{CE})$   $\triangleright$  backpropagation, minimize  $\ell$ 
19:   end for
20: end for
21:  $\mathcal{V} = \mathcal{C}(\mathcal{W}_{Te}, \vec{\Theta})$   $\triangleright$  predictions on test set
22: return( $\mathcal{V}$ )

```

window to be classified is highlighted in black, overlapping windows excluded from the analysis are marked in white, and windows that enrich the temporal-frequency picture around the central window are shown in gray. It is important to note that the overlap is only used for generating a larger dataset for training, effectively reducing the dependence on the initial index from which the Mel spectrogram calculation begins.

TABLE II

TRANSFORMER PARAMETER VALUES

Parameter	Value
Manufacturer / Connection	Elin, 1968 / Yy05/d5
Voltage [kV] / Power [MVA]	220/115/10.5 / 150/150/50
Tap Changer	Motor, MLRG 3/720, 27 pos.

IV. CASE STUDY

To experimentally validate the previous considerations, tests were conducted on an OLTC that was mounted on a transformer, named ELIN, which has been out of operation for a long period. The transformer is a three-winding regulating autotransformer RTdXP 160000/200, as shown in Fig. 6(a) with the characteristics displayed in Table II.

Fig. 6(b) illustrates the OLTC control system for voltage regulation. The line drop compensator calculates the voltage drop to ensure a stable output voltage. The system uses a feedback loop to compare the measured voltage with the set voltage and then operates the motor and control logic to adjust the transformer taps via the tap selector and diverter switch. To avoid reacting to minor voltage fluctuations, a deadband and time delay are employed for stable regulation.

Fig. 6(c) shows the recorded audio OLTC fingerprint that corresponds with one OLTC transition. One way to perform signal segmentation is empirically based on setting thresholds that are based on certain statistical characteristics of the signal, such as standard deviation, mean value, or signal energy.

However, this approach has proven impractical because it requires significant human interaction, and it is sensitive to changes and noise. For that reason, the segmentation of the signal involved a comprehensive approach where expert knowledge was embedded to guide the process. Iterative corrections were essential to fine-tune the segmentation, ensuring the accuracy and reliability of the results. The stages of the segmentation process were defined as displayed in Table III.

The main difference between the signal in Fig. 6(c) and the one in [4] is in segment #7. This segment does not appear in the vibration fingerprint, but, here, it stands out due to its notable variation in the audio signal.

The described segmentation process has now been performed on the entire audio recording, lasting approximately 8.5 million samples, recorded at a sampling rate of

TABLE III
SEGMENTATION OF TRANSFORMER OLTC STATES

State	Description
#1	Idle state and motor operation
#2	Motor starts
#3	Geneva drive activation
#4	Tap selector stops and end of Geneva drive
#5	Diverter switch operation
#6	Motor stops
#7	Events associated with the mechanical braking process

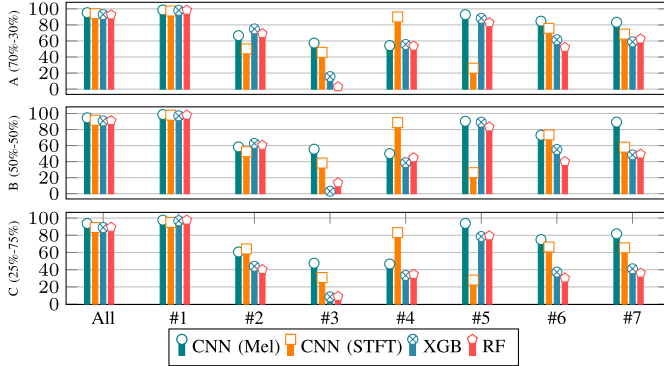


Fig. 7. Classification accuracy for basic models with different training–testing proportions (A, B, and C).

48 kHz using a commercial USB-based audio card, for all 27 tap transitions of the OLTC. For sound recording, the Proel DM800 microphone (60-Hz–14-kHz frequency range, –53-dB sensitivity, and 600-Ω impedance) was used. The recorded audio was processed via the E-MU 0404 USB interface, which provides 24-bit depth and a sampling rate of up to 192 kHz. The segmented data were further utilized in the supervised deep learning process.

V. EXPERIMENTAL RESULTS

A. Benchmarking Classifiers

1) *Basic Classifiers*: Fig. 7 displays the classification accuracy results for the basic classifiers for different train–test configurations (A: 70%–30%; B: 50%–50%; and C: 25%–75%).

It can be observed that the best results are obtained with the CNN model with the Mel spectrogram as input. There are specific cases where other models perform better, e.g., state #4 is best captured with CNN with STFT as input and state #2 is best captured with the XGB algorithm.

2) *Advanced Classifiers*: Taking the best model in Fig. 7, i.e., CNN with Mel spectrogram, a probabilistic classification model has been implemented via Monte Carlo dropout (see Section III-A1). In addition, PCA-based feature extraction methods have been implemented within the CNN model. Fig. 8 displays the obtained results for the same three train–test proportions, as shown in Fig. 7.

The best overall results are obtained for PCA (CNN), except for the configuration with the least training data (configuration C). This partially supports that under conditions of uncertainty, probabilistic classification may provide a useful representation of this information and support decision-making. As for

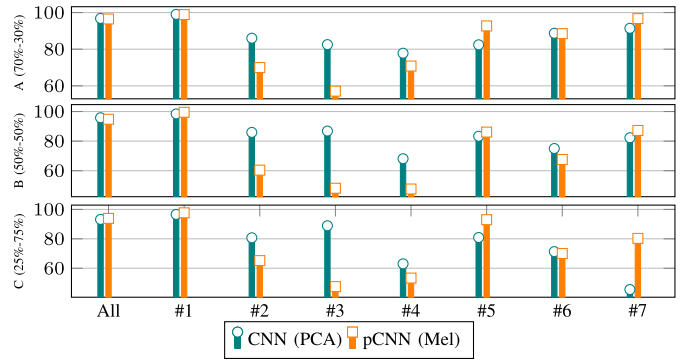


Fig. 8. Classification accuracy for advanced models with different training–testing proportions (A, B, and C).

specific states, #1, #5, and #7 are best captured with the probabilistic CNN model, and the remainder of states are best captured with the CNN (PCA) model.

Figs. 7 and 8 present a comparative analysis of various predictive models, their operational states, and different training and testing proportions. This analysis underscores a core observation: augmenting the training data improves the overall accuracy of the classifiers. However, despite this general trend, the improvement in accuracy for individual categories is marginal. This suggests that simply expanding the dataset size might not adequately meet the precision requirements for each category.

3) *Robustness of Classifiers*: The classification accuracy of different models has been further tested to evaluate their robustness with noise. To this end, the signal-to-noise (SNR) ratio [dB] has been predefined [31]

$$\text{SNR} = 10 \log \frac{\text{rms}_{\text{signal}}^2}{\text{rms}_{\text{noise}}^2} \quad (4)$$

where rms is the root mean square operation.

With predefined SNR levels, the corresponding noise characteristics have been calculated from (4), and this has been used to generate white Gaussian noise with the corresponding standard deviation. Fig. 9 shows the classification accuracy for different models, including different levels of noise.

It can be observed from Fig. 9 that with higher SNR levels (10 and 5 dB), CNN with PCA performs the best. It can be observed that for specific states, in the 10-dB case, CNN (PCA) has better accuracy for three states (#2, #3, and #6) and pCNN as well (#5, #6, and #7). In the 5-dB case, CNN (PCA) has better accuracy for one state (#6) while pCNN for four states (#2, #3, #5, and #7). For 0 dB, the best model is pCNN, which obtains the best accuracy for five states (#1, #3, and #5–#7).

For lower levels, it can be observed that CNN with Mel spectrograms becomes good only for specific states (#2, #3, and #5) and is not able to detect other states. The CNN (PCA) performs in a robust way consistently detecting states #5 and #6. Note that for the lowest level, the RF performs best, due to its capacity to detect state #1, which has the highest number of samples (see Section VI-B).

4) *Classifier Complexity*: Table IV summarizes the complexity of the models trained in the experiments by showing the total number of parameters of the CNN and the number of

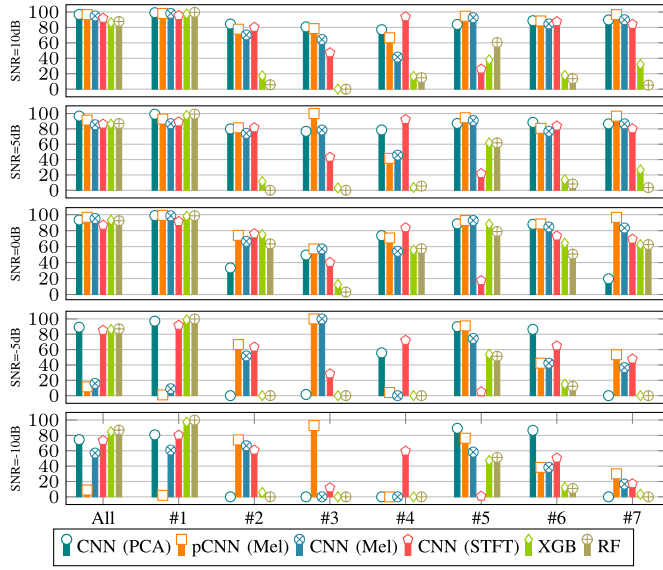


Fig. 9. Classification accuracy with respect to different noise levels.

TABLE IV
MODEL COMPLEXITY SUMMARY

Model	# param.	# mult. + add.	# activ.
CNN (PCA)	29,403,067	36,752,675	132
CNN (Mel)	1,420,215	118,112,634	688,189
CNN (STFT)	5,401,527	472,477,050	2,755,645

multiplication, additions, and activation function calls required to calculate one inference.

The input data preprocessing is also different for each model.

- 1) The CNN (PCA) gets a 294×100 matrix obtained from calculating the PCA scores from the STFT and concatenating these scores for successive time instants (see Section III-A3). There is an initial PCA processing cost to calculate and select the principal components, but, thereafter, the only cost at inference is an additional matrix operation to calculate the PCA scores. Nevertheless, the initial PCA computation cost is quite heavy as it takes all captured samples, groups them by the annotated states, and then calculates the PCA for each state. Therefore, the time and memory requirements increase with $O(V^3)$, where V is the number of PCA variables.
- 2) The CNN (Mel) gets a 128×96 matrix obtained from a Mel spectrogram and reduces the size to the previous dimensions.
- 3) The CNN (STFT) gets a 513×96 matrix obtained by selecting 96 consecutive time instants from the STFT.

In terms of balancing the model efficacy and complexity, the PCA-based approach emerges as the most effective, yet complex model. Consequently, subsequent research has shifted focus toward employing the Mel spectrogram technique.

B. Final Model

After selecting the best classification architecture following the proposed approach (see Fig. 1), first, the proposed extended time window approach is implemented evaluating

TABLE V
ACCURACY FOR EXTENDED TIME-WINDOW VALUES. BEST RESULTS HIGHLIGHTED (EXCLUDING $n = 8^*$)

n	All	State-specific						
		#1	#2	#3	#4	#5	#6	#7
4	96.4	98.5	83.2	61.7	96.8	87.9	97.4	89.8
5	97.0	98.6	85.7	67.0	99.5	95.2	94.8	88.1
6	97.5	99.0	87.4	76.6	99.1	95.2	92.2	89.0
7	97.7	98.9	89.9	79.8	99.1	96.0	94.8	89.8
8	97.6	99.0	85.7	85.1	99.0	96.0	88.7	89.8
8*	98.0	99.0	91.7	76.6	98.6	96.8	93.9	99.1

Legend: 8*: corrected mislabeling (see Section VI-A: Accuracy & Labelling Challenges).

the sensitivity of the time window; then, a computationally efficient approach is implemented; and finally, the proposed approach is benchmarked against state-of-the-art methods.

1) *Sensitivity Assessment of the Time Window*: For the sensitivity assessment, all the configurations have been tested with the Mel spectrogram with an equitable data division of 50%–50% for training and testing purposes. The preprocessing phase is initiated by computing the Mel spectrogram, with $M = 128$, $w = 1024$, and $o = 75\%$. The time–frequency matrix, D , is then analyzed to classify the spectral content into predefined categories (see Table III). Through the application of Algorithm 1, with mode = “horizontal,” Table V displays the outcomes of this method. A sensitivity assessment of the number of considered windows has been performed, ranging from $n = 4$ to $n = 8$ (with $d = 4$).

Results indicate that the classification accuracy generally improves as the number of windows increases. Specifically, at $n = 8$, the total window span is of 17×1024 samples, covering approximately 360 ms (sampling rate of 48 kHz), with an input image of 17×128 pixels. The obtained accuracy is greater than the accuracy of each classification approach when employing a 50/50 training–validation split (see Figs. 7 and 8). Also, each class in this case has classification accuracy above 85%.

2) *Input Data Configuration*: The achieved accuracy results validate the proposed classification approach (see Table V). This accuracy was attained by observing a time span of approximately 350 ms. Moving forward, for improved computational efficiency, the strategy involves preserving this temporal framework while seeking to streamline the dataset.

This can be achieved by reducing the number of windows, with a focus on retaining the pivotal central window to ensure the integrity of the classification process. The initial effort to reduce data volume involves selecting a subset of windows, specifically choosing even or odd windows from the auxiliary windows. In this case, based on the best configuration, $n = 8$, eight auxiliary windows surround the core window. To adapt this strategy to the CNN architecture, the windows are organized into a suitable 1-D format for the input layer. This approach significantly lowers computational requirements by focusing on temporal, rather than spatial, relationships between windows, bypassing the computational complexity, and interwindow dependencies associated with

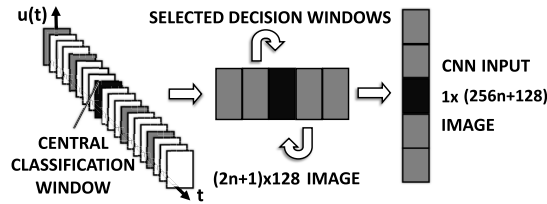


Fig. 10. Vertical arrangement of windows.

TABLE VI
PREDICTION ACCURACY FOR WINDOWING STRATEGIES

Exp.	Model	All	State-specific						
			#1	#2	#3	#4	#5	#6	#7
Vert.	#8a	98.4	98.7	95.9	90.4	99.1	96.8	97.4	100
	#8b	98.0	98.7	96.7	76.6	98.2	98.4	95.7	97.2
	#8c	97.4	98.7	90.9	66.0	97.7	96.8	94.8	98.1
	#8d	97.1	98.2	94.2	64.9	99.1	96.8	94.8	96.3
Size	#8e	98.5	99.1	95.0	92.6	98.6	96.8	92.2	99.1
	#8f	98.2	98.5	94.2	90.4	99.1	97.6	98.3	99.1
	#8g	98.1	98.3	95.9	87.2	99.5	97.6	98.3	99.1

Legend: Exp.: Experiment; Vert.: Vertical window arrangement; #8a: ext.8x #8b: ext.8x-sel.even wnds; #8b: ext.8x-sel.even wnds; #8c: ext.8x-sel.odd wnds; #8d: ext.8x-4 side wnds and central wnd selected. Size: 1D-CNN size #8e: ext.8x, CNN: 16-32-64, FC: 128 #8f: ext.8x, CNN: 16-32-64, FC: 64 #8g: ext.8x (reduced input), CNN: 8-16-32, FC: 128. FC: Fully Connected.

2-D convolution. Fig. 10 shows the vertical arrangement of windows, which makes possible the 1-D processing.

Through the application of Algorithm 1, with mode = "vertical," Table VI displays the outcomes of these experiments. The experiments with vertical alignment windows include extended vertical windows with $n = 8$ for (#8a) odd and even windows, (#8b) even windows, (#8c) odd windows, and (#8d) the two outermost windows on either side of the central window.

The results show that with a vertical arrangement of windows, a higher accuracy is achieved compared to other time–frequency processing methods (see Table V). The best accuracy was obtained when all 17 windows were utilized in the decision-making process (configuration #8a). The reduction of the input data by decreasing the number of windows led to a decrease in accuracy, notably in state #3, where accuracy dropped below 80%. This decline in accuracy indicates that reducing the number of windows is not an efficient solution to streamline the dataset. Taking the best model from vertical alignment tests, i.e., configuration #8a, in order to reduce the complexity of the CNN model, different tests have been performed (see Table VI, size row). Results show that the best outcome was achieved with a configuration of 16, 32, and 64 CNN layers plus a fully connected layer with 128 neurons, with an overall accuracy of 98.5%. Finally, it is observed that lowering the number of neurons in the fully connected layer decreased accuracy.

3) *Benchmarking With Ensemble of Classifiers via AutoML:* AutoML models have shown promising evidence for an automated combination of machine learning models for classification and regression tasks. In this section, the H2O AutoML algorithm is implemented [32], a scalable automatic machine learning framework that efficiently evaluates a range

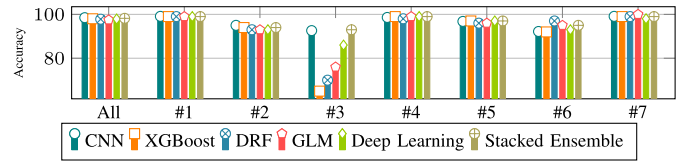


Fig. 11. Accuracy comparison between CNN and AutoML models across categories.

of models. After adapting the mel spectrograms to the H2O format, different base classifiers including NNs in fully connected multilayer perceptron configurations, XGB, generalized linear model (GLM), distributed RF (DRF), and stacking-based combination of the base classifier were trained and tuned, resulting in a total of 20 different models. The best-performing model is a stacked combination of NN, XGB, GLM, and DRF, combined through a GLM algorithm acting as a metalearner. Fig. 11 displays the results obtained for the individual base classifiers and the stacked ensemble model.

It can be seen that the developed CNN model outperforms the best AutoML model, which is observable, especially in category #3. Only the stacked ensemble model matches the CNN model's performance. However, this ensemble model combines four different classifiers through a dedicated metalearner model, which adds substantial complexity to the final classifier, without sufficient benefit over the CNN model.

C. Application

The ELIN transformer tested in this study was decommissioned and, thus, could not be put online. Accordingly, a sister active transformer was used to assess the operation conditions of an active transformer.

1) *Active Transformer:* Humming noise signals were recorded from the active transformer with microphones placed at the same distance as in the ELIN setup. These recordings were then combined with the original OLTC signals to create a synthetic composite signal including humming noise and OLTC-produced audio signatures. In this scenario, two different experiments were performed: 1) direct application of the proposed classification approach and 2) filter noise and then apply the proposed approach.

In the second experiment, an STFT-based DNN denoising framework described in [6] was used, which takes as input the acquired signals and the synthetic composite signal. To achieve robustness in the training data, the noise signal amplitude was varied by $\pm 10\%$ around the recorded amplitude and added to the OLTC signals. The DNN was trained to produce STFT masks that eliminate noise in the STFT domain. The trained DNN was then used for signal denoising by applying the inverse STFT to the product of the estimated mask and the STFT of the synthetic composite signal.

Fig. 12 shows the obtained results. The first experiment with the direct application of the proposed OLTC operation monitoring framework failed to classify states effectively using the synthetic composite signal [see Fig. 12 (top)]. However, for the denoising network followed by the proposed classification network, the results were satisfactory [see Fig. 12 (bottom)]. It can be observed that all target states of the cleaned signal

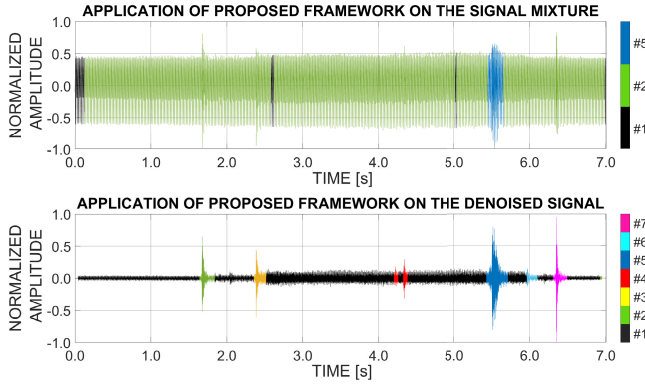


Fig. 12. Classification results of the proposed approach in an active transformer. Top: Classification without denoising. Bottom: Classification after denoising humming noise through DNN masking; for state definitions, see Table III.

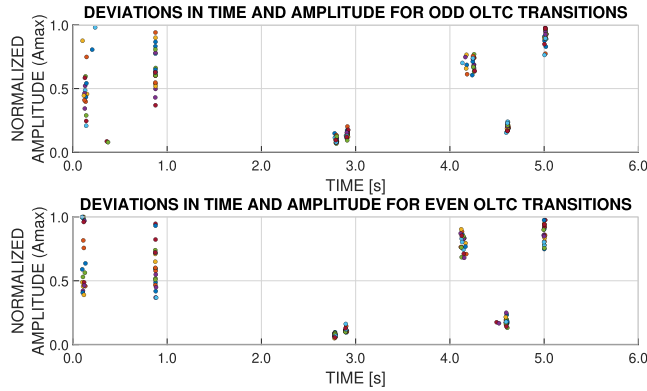


Fig. 13. Extracted local peaks for odd and even tap transitions.

that corresponds to one OLTC tap transition are accurately labeled. The integration of denoising and segmentation into a single unified procedure, significantly reduces complexity and improves the accuracy of diagnostic information extraction.

2) *Operation Monitoring*: Automatic signal segmentation simplifies the extraction of diagnostic features. As demonstrated in [4], the distances and amplitudes of local peaks in the envelope of the acoustic signature are important diagnostic parameters. To extract these features effectively, it is essential to focus on specific segments of the signal. Furthermore, signal segmentation aids in removing undesired content from the signal. For instance, additional noise observed between the segments labeled as Geneva drive activation and diverter switch operation can be effectively eliminated by applying the least mean squares (LMS) adaptive filter specifically to this part of the signal (see Appendix B for more details). By removing this noise through LMS, it becomes possible to extract the local peaks in the envelope signal that corresponds to the states when the tap selector stops and the end of the Geneva drive operation.

Fig. 13 shows the extracted peaks for odd and even tap transitions from 81 signatures captured for various OLTC transitions (27 up, 27 down, and 27 up operations).

From Fig. 13, it can be observed that the proposed approach significantly simplifies the detection of deviations in time and amplitude of acoustic signatures.

TABLE VII
CONFUSION MATRIX, TESTED ON THE VALIDATION SET

Actual	Predicted						
	#1	#2	#3	#4	#5	#6	#7
#1	3140	3	7	4	2	5	4
#2	15	101	3	0	0	0	0
#3	15	0	79	0	0	0	0
#4	4	0	0	218	0	0	0
#5	6	0	0	0	118	0	0
#6	11	2	0	0	0	102	0
#7	4	9	0	0	0	0	105

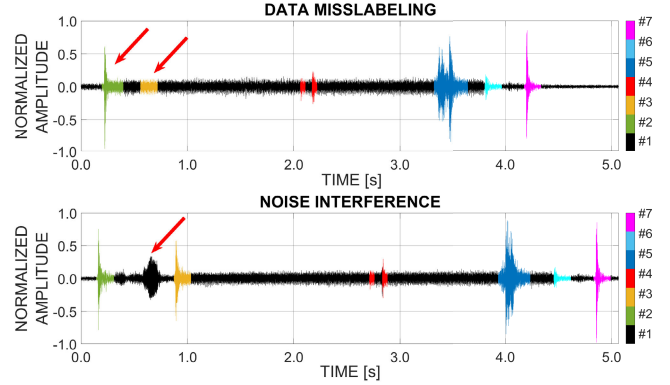


Fig. 14. Labeling errors and noise interference. For state definitions, see Table III.

VI. DISCUSSION

The final objective of the proposed approach is to detect the operation status of the OLTC through audio signals. In this manner, maintenance management teams will be able to accurately monitor the correct and reliable operation of transformers in a cost-effective way. However, the cost-effectiveness also includes other challenges discussed in this section.

A. Accuracy and Labeling Challenges

To ascertain the sources of the residual errors, constituting less than 3% of the validation dataset for the $n = 8$ case (see Table V), an analysis was conducted using the confusion matrix specific to this scenario, as displayed in Table VII.

Transitional windows, representing shifts between different states, emerge as the primary source of errors in the validation dataset, as highlighted by the confusion matrix. These boundary windows, where the classification becomes challenging due to the nearly equal distribution of samples across categories, predominantly account for misclassifications. However, the most significant nontransitional misclassification observed involves nine windows from state #7 being erroneously classified as state #2. This particular instance underscores the challenges in distinguishing between these specific categories under certain conditions, and it represents the largest error within the analyzed dataset.

This was identified through a meticulous analysis, particularly focusing on the sample range 4.55×10^6 – 4.58×10^6 . Fig. 14 shows the signal within this segment and indicates a clear case of mislabelling, contributing to the validation errors.

This scenario underscores the inherent challenges in data labeling, even with rigorous expert-based manual scrutiny.

TABLE VIII
DATASET CLASS-BY-CLASS SAMPLE DISTRIBUTION

Balancing	State-specific						
	#1	#2	#3	#4	#5	#6	#7
No	83.2	3.0	1.6	5.0	2.4	2.2	2.6
Undersampling	22.9	13.9	7.5	22.9	10.9	10.0	11.9
Oversampling	14.3	14.3	14.3	14.3	14.3	14.3	14.3

Conversely, this also highlights the utility of error detection via CNN, which is a key tool for enhancing the accuracy and reliability of data labeling protocols, thereby refining the overall quality of the dataset. Following the detection of this error, the network was retested with the same $n = 8$ configuration, and the results obtained are shown in Table V, named $n = 8^*$ configuration.

The additional errors, observed in two characteristic OLTC fingerprints, in the validation dataset can be attributed to noise interference (see Fig. 14). Upon listening to the audio recordings, it became clear that this noise was caused by the human voice. This highlights the importance of ensuring a quiet environment for audio-based monitoring to capture high-quality signals. Detecting such anomalies can be useful to discard these recordings or subject them to further analysis.

B. Sample Balancing and Testing Strategies

The achieved overall accuracy through vertical window arrangement can be considered satisfactory (98.5%; see Table VI). However, an imbalance in accuracy across different states has been observed. Among the OLTC states, it is notable that state #1 encompasses a significantly larger volume of data in comparison to the other states. Such a distribution pattern was anticipated, given that state #1 encompasses inactive periods of the OLTC, or when the motor is operational, among various scenarios. The distribution of data by states is shown in Table VIII. It is likely that this data imbalance affects the learning ability of the model, particularly for states with fewer samples.

To evaluate the learning ability of the model under different sampling scenarios, two sample balancing techniques, undersampling and oversampling, have been applied. In the undersampling strategy, data from state #1 are randomly sampled until its dimension is aligned with state #4, which has the second largest amount of samples. With the oversampling approach, the number of instances in categories other than #1 was artificially increased to achieve numerical parity. Each artificially generated sample is produced by adding a small amount of random noise (less than 0.1%) to the original data. Table VIII shows data distribution following the implementation of two data-balancing strategies.

Previously, overlap was used to increase the training set and reduce dependence on the initial index from which the Mel spectrogram is computed, as demonstrated in Fig. 6. Similarly, it is possible to form training and validation data using a nonoverlapping Mel spectrogram by shifting iteratively in the time domain. This procedure requires adjusting the start position in the time domain to calculate the Mel spectrogram during each iteration. The distance for each shift is determined

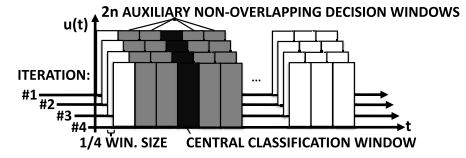


Fig. 15. No overlapping train/test approach—four iterations.

TABLE IX
PREDICTION ACCURACY FOR SAMPLE BALANCING MODELS

Model	All	State-specific						
		#1	#2	#3	#4	#5	#6	#7
CNN1	98.1	98.0	96.4	96.8	100	98.1	97.3	99.0
CNN2	97.8	97.7	98.0	93.1	100	98.5	98.2	99.3
CNN3	98.0	97.9	97.1	96.7	99.9	99.6	97.7	99.5
Comb.	98.2	98.1	97.0	97.2	100	98.6	97.9	99.2

by the ratio of the total window size to the number of iterations. Fig. 15 illustrates this approach, employing four surrounding windows adjacent to the central window that is subject to classification.

Accordingly, to evaluate different data sampling strategies (see Table VIII) without overlap, four CNN models are designed: (CNN1) no balance; (CNN2) undersampling; (CNN3) oversampling; and (Comb.) combination of CNN1, CNN2, and CNN3. This combination is done through an ensemble approach where the decision-making process for categorization follows a majority voting scheme from the three networks.

The evaluation employed a testing methodology that involved repeated iterations, with the total number of iterations equalling the window size and an iteration step of 1. This iterative testing approach reduces memory requirements.

During the testing phase, for each iteration, a confusion matrix is calculated for each network and their combination. The confusion matrix results are cumulatively added to the final confusion matrices. Finally, the accuracy of each category and the overall accuracy of all calculated matrices are computed. Table IX summarizes the accuracy performance results.

It can be concluded that the CNN models exhibit high overall predictive accuracy on the validation set. The ensemble model slightly outperforms the individual models, indicating robust classification capabilities across different states. As for the effectiveness of data-balancing strategies, it is observed that the combined model marginally improves the performance of individual models across all state-specific datasets.

C. Relevance of the Proposed Solution

For rapid audio-based diagnostics of OLTCs, it is essential to ensure efficient signal recording, often relying on commercial audio systems. Such systems are typically limited by the absence of additional inputs for triggering measurements or auxiliary channels for parameters such as current monitoring. This article provides a framework for software development that enables rapid OLTC diagnostics based on audio signals. The proposed approach simplifies the extraction of diagnostic features from specific segments of the signal, which are critical for analyzing OLTC conditions.

Furthermore, by enabling the comparison of individual signal segments, the framework facilitates the analysis of signals recorded at different time intervals or from various OLTCs of the same type, thereby enhancing fault detection and trend analysis. The method demonstrates robustness by effectively handling audio signals recorded during both active and inactive transformer operations, ensuring adaptability to diverse real-world scenarios.

D. Experimental Insights

The spatial configuration of the OLTC depends on its constituent components. Accordingly, audio-based monitoring solutions are tailored to specific OLTC types. To guide possible future configurations from our experimental experience, this section discusses the main experimental insights.

The highest SNR for the diverter switch was achieved when the microphone was positioned on the OLTC cover. This is because the cover is thinnest at the top, allowing less dampening of audio signals. However, this position suppresses signals from other components, making it suitable only for isolated monitoring of the diverter switch.

When the microphone was too close to the motor, the motor noise masked diagnostic signals from other components, emphasizing the need to avoid such placements. As for the tap selector, the optimal position was found to be near the middle of the OLTC housing because the signals at the top or bottom were heavily dampened.

Finally, audio signals collected by microphones showed an advantage over accelerometers because sound waves travel more effectively through air than through solid materials. With a proper microphone placement, it may be possible to capture signals from multiple sources more effectively than with accelerometers.

E. Limitations

The objective of this research work is to develop an OLTC operation monitoring framework based on audio signals. Audio signals are prone to noise and interference, which also impact the proposed approach. This has been tested in this work with the evaluation of the robustness of classifiers with artificial noise and humming noise emerging from active transformers. In the first case, the classification performance was evaluated directly with the classifier, and in the second case, a noise filter was applied to eliminate noise in the frequency domain. In realistic application contexts, transformers may be placed in different locations and surrounded by diverse sources of noise, and this will need to be thoroughly integrated into the framework.

As for the experimental setup, audio-based diagnostics require that the microphone is positioned in the same location for repeated measurements to avoid signal delays, ensuring consistency in data collection and analysis. This is a relevant requirement for the implementation of real systems.

The proposed approach relies on manual labeling of OLTC states, which is a tedious task not scalable for online systems. In this direction, automatic labeling and classification alternatives must be explored through novel machine learning solutions, such as self-supervised audio solutions [33].

VII. CONCLUSION

Transformers are key assets for reliable grid operation. The OLTC is a key transformer mechanism, which enables automatic adaptation to different grid voltage levels and contributes to grid stability. However, repeated operation of the OLTC accelerates aging and influences the reliability and efficiency of the transformer. Continuous monitoring of the operational status of the OLTC is an interesting technological solution, as it will enable detecting deviations from normal operation and raising alarms.

Consequently, this research presents noninvasive and cost-effective solutions to monitor the operational status of the OLTC system through audio signals. The proposed approach enables broad applicability through widely available commercial audio systems and offers an additional mechanism to improve the reliability of the electrical grid. This has been achieved through a novel acoustic signal-based classification method based on signal processing and machine learning methods.

The framework is designed through an SPL strategy, which first learns coarse classification criteria, and subsequently, it is further refined through signal processing and CNN methods. In the initial phase, different methods are evaluated with respect to accuracy, complexity, and robustness to noise. After selecting the best tradeoff architecture, a fit-for-purpose signal processing algorithm is designed, which is subsequently coupled with the CNN model. The proposed approach has been compared with AutoML-based ensemble solutions. The obtained results show that the proposed strategy is more accurate and efficient. Finally, a detailed discussion is provided with respect to labeling challenges, computational efficiency, and data-balancing strategies.

The proposed methodology was tested on a transformer OLTC, and it showed high accuracy. This method contributes to the development of audio-based diagnostic tools, which can serve as a powerful tool for a rapid and efficient assessment of OLTC conditions. Future work may address the development of unsupervised and self-supervised learning strategies to alleviate the OLTC labeling process.

APPENDIX A SELF-PACED LEARNING

For a given supervised learning problem, with a dataset $D = \{x_i, y_i\}_{i=1}^n$, $x_i \in X$, $y_i \in Y$, $X \in \mathbb{R}^{n \times f}$ for f features, and $Y \in \mathbb{R}^{n \times 1}$, and learning algorithm fitting parameters, w , the SPL approach is defined as follows [14], [15]:

$$\mathcal{L} = r(w) + \sum_{i=1}^n v_i f(x_i, y_i, w) - \lambda \sum_{i=1}^n v_i \quad (5)$$

where w denotes the model weights, $f(\cdot)$ is the loss function evaluated in (x_i, y_i, w) , λ is the threshold variable, and v is a binary variable, which decides the training dynamics.

SPL is a two-step learning process in which $\mathcal{L}$ is trained twice while keeping variables fixed. Namely, in the first step, $\mathcal{L}$ is optimized by evaluating $f(\cdot)$ and λ , setting $v = 1$ if $f(\cdot) < \lambda$ and 0 otherwise. This step decides the complexity of the training. Once fixed v , the second step optimizes w by backpropagation.

APPENDIX B

LEAST MEAN SQUARES FILTER

The applied LMS filter is defined as follows:

$$\text{Filter output: } y[n] = \mathbf{w}[n]^T \mathbf{x}[n]$$

$$\text{Error: } e[n] = d[n] - y[n]$$

$$\text{Weight update: } \mathbf{w}[n+1] = \mathbf{w}[n] + \mu \mathbf{x}[n] e[n]$$

where $\mathbf{w}[n]$ is the weight vector, $\mathbf{x}[n]$ is the input signal, $d[n]$ is the desired signal, and μ is the step size.

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